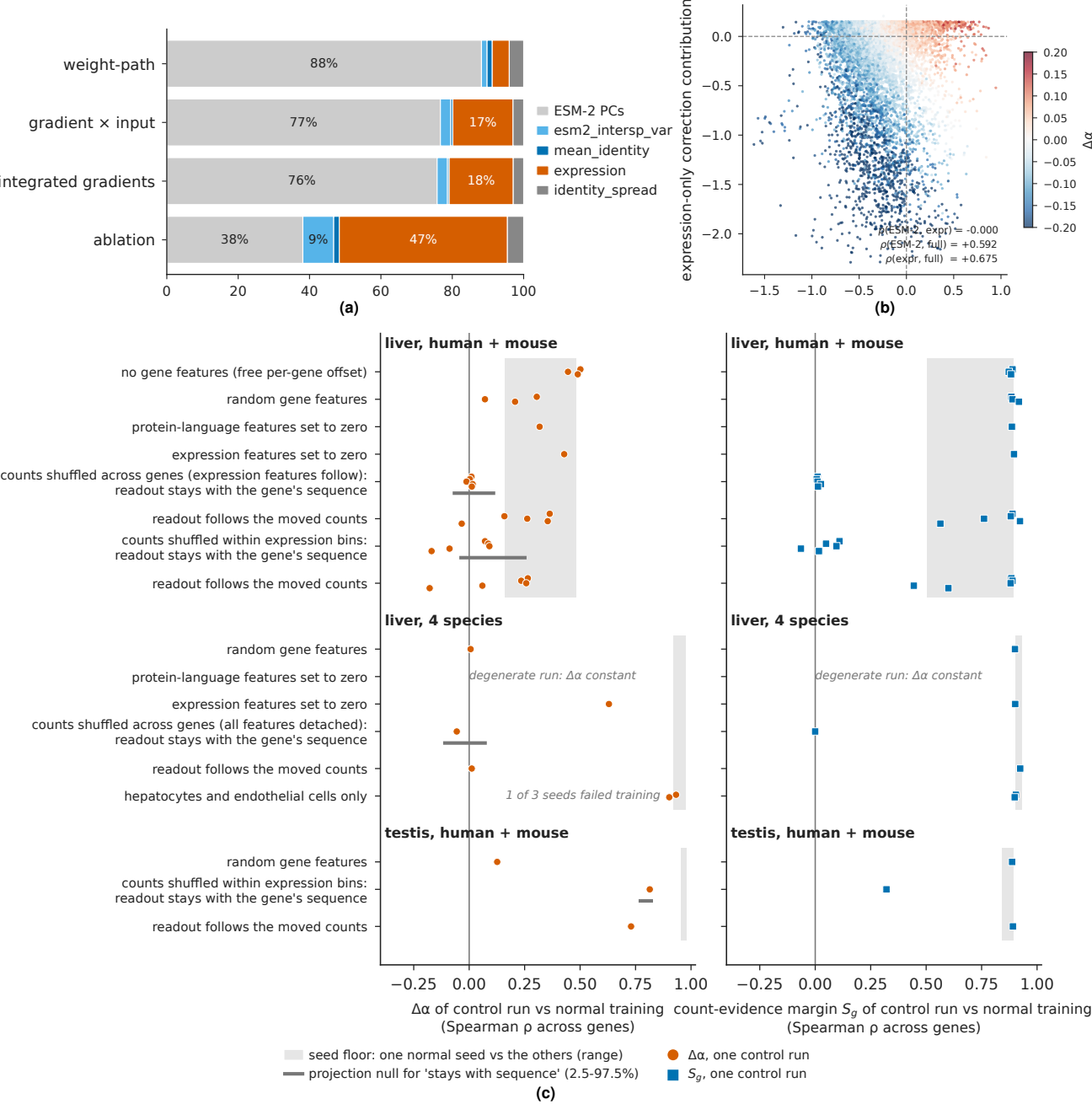

weight-path
gradient × input
integrated gradients
ablation
88%
77%
17%
76%
18%
38%
9%
47%
0
20
40
60
80
100
ESM-2 PCs
esm2_intersp_var
mean_identity
expression
identity_spread
(a)
expression-only correction contribution
0.0
−0.5
−1.0
−1.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
ρ(ESM-2, expr) = -0.000
ρ(ESM-2, full) = +0.592
ρ(expr, full) = +0.675
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Δα
(b)
liver, human + mouse
no gene features (free per-gene offset)
random gene features
protein-language features set to zero
expression features set to zero
counts shuffled across genes (expression features follow): readout stays with the gene's sequence
readout follows the moved counts
counts shuffled within expression bins: readout stays with the gene's sequence
readout follows the moved counts
liver, 4 species
random gene features
protein-language features set to zero
expression features set to zero
counts shuffled across genes (all features detached): readout stays with the gene's sequence
readout follows the moved counts
hepatocytes and endothelial cells only
degenerate run: Δα constant
1 of 3 seeds failed training
testis, human + mouse
random gene features
counts shuffled within expression bins: readout stays with the gene's sequence
readout follows the moved counts
liver, human + mouse
liver, 4 species
testis, human + mouse
degenerate run: Δα constant
−0.25
0.00
0.25
0.50
0.75
1.00
Δα of control run vs normal training (Spearman ρ across genes)
count-evidence margin S_g of control run vs normal training (Spearman ρ across genes)
seed floor: one normal seed vs the others (range)
projection null for 'stays with sequence' (2.5-97.5%)
Δα, one control run
S_g, one control run
(c)